

Precalculus Placement Self-Assessment

Universal Grading Rubric — Randomized Version

Instructions for graders: This rubric applies regardless of the order questions appear or the specific numbers used. Match each question to its *topic*, then apply the criteria below. Total points: **50**. Partial credit is awarded as indicated.

Score Summary

#	Topic	Pts	Earned
1	Exponent Rules	2	
2	Number Classification	4	
3	Polynomial Subtraction	1	
4	Rational Expression Subtraction	1	
5	Factoring Quadratics	1	
6	Modeling with Variables	1	
7	Absolute Value Inequalities	2	
8	Distance & Midpoint	3	
9	Linear Functions from Graphs	2	
10	Domain & Range of Square Root	2	
11	Table of Values & Graphing	7	
12	Net Change & Average Rate of Change	2	
13	Function Transformations	1	
14	Function Composition	2	
15	Inverse Functions from a Table	1	
16	Local Extrema of Polynomials	2	
17	Polynomial Long/Synthetic Division	1	
18	Polynomial Inequalities	1	
19	Exponential Transformations	4	
20	Logarithm $\leftrightarrow$ Exponential Form	2	
21	Laws of Logarithms	1	
22	Inverse Trigonometry	1	
23	Sine Function Transformations	3	
24	Period of Cotangent	2	
25	Vector Representation	1	
TOTAL		50	

Q1 | Exponent Rules (SPreCalc8M 1.2)

Points: 2 pts

Part (a)

- 1 pt — Correctly subtracts exponents when dividing same-base terms ($x^m / x^n = x^{(m-n)}$).
- 1 pt — Moves any factor with a negative exponent to the opposite position (numerator↔denominator) and makes the exponent positive.
- Award 0 if student adds exponents instead of subtracting, or leaves negative exponents in the answer.

Part (b)

- Credit included in the 2 pts above — award full credit only if the outer exponent is distributed correctly to EVERY factor (numerator AND denominator), and numeric coefficients are evaluated (e.g., $2^3 = 8$).

Common Errors (deduct points):

- Adding exponents when dividing ($x^2 / x^3 \rightarrow x^5$ instead of x^{-1}).
- Forgetting to raise the coefficient to the outer power ($(a^3/2b^3)^3 \rightarrow a^9/2b^9$ instead of $a^9/8b^9$).
- Leaving negative exponents in the final answer.

Q2 | Number Classification (SPreCalc8M 1.1)

Points: 4 pts

Part (a – Natural number)

- 1 pt — Any positive counting integer (1, 2, 3, ...).

Part (b – Integer, not natural)

- 1 pt — Zero or any negative integer (... , -3, -2, -1, 0). Must NOT be a natural number.

Part (c – Rational, not integer)

- 1 pt — Any fraction p/q ($q \neq 0$) that does not simplify to a whole number (e.g., $1/2$, $-3/4$, 0.7).

Part (d – Irrational number)

- 1 pt — A non-terminating, non-repeating decimal (π , e , $\sqrt{2}$, $\sqrt{3}$, etc.). Accept any correct irrational; deduct if student writes a rational approximation like 3.14 and claims it is irrational.

Common Errors (deduct points):

- Writing a natural number for (b) or (c).
- Writing a terminating decimal for (d).
- Writing a rational approximation of π (3.14) and calling it irrational — deduct that point.

Q3 | Polynomial Subtraction (SPreCalc8M 1.3)

Points: 1 pt

Correct Method:

- Distribute the negative sign across the entire second polynomial before combining.
- Combine like terms correctly (x^2 , x , and constant terms separately).
- 1 pt for a fully correct simplified answer; 0 pts if the negative sign is not distributed (sign error on any term).

Common Errors (deduct points):

- Subtracting only the first term of the second polynomial and leaving others unchanged.
- Arithmetic error in combining like terms after distributing correctly — deduct the 1 pt.

Q4 | Rational Expression Subtraction (SPreCalc8M 1.4)

Points: 1 pt

Correct Method:

- Convert the whole number to a fraction with the same denominator as the rational expression.
- Subtract numerators correctly (distribute the negative into the whole-number numerator).
- Simplify the resulting numerator; leave denominator factored or expanded consistently.
- 1 pt for a correct fully simplified result; 0 if common denominator is wrong or numerator arithmetic is incorrect.

Common Errors (deduct points):

- Forgetting to multiply the integer by the denominator when rewriting it as a fraction.
- Distributing the subtraction incorrectly into the multiplied terms.

Q5 | Factoring Quadratics (SPreCalc8M 1.5)

Points: 1 pt

Correct Method:

- Identify two numbers that multiply to the constant term (c) and add to the linear coefficient (b).
- Write the factored form correctly.
- Set each factor equal to zero and solve for x.
- 1 pt for both correct roots; 0 if factoring is wrong or only one root is found without justification.

Common Errors (deduct points):

- Sign errors when choosing factor pairs.
- Writing the factors but forgetting to solve for x (stops before stating the roots).
- Using the quadratic formula instead of factoring — award the point if both roots are correct, but note the method.

Q6 | Modeling with Variables (SPreCalc8M 1.7)

Points: 1 pt

Correct Method:

- Recognize the structure: Total = (rate) $\times$ (quantity).
- Express the total as (dollar amount per month) $\times$ n, where n is the number of months.
- 1 pt for a correct algebraic expression using the given rate and the defined variable.

Common Errors (deduct points):

- Writing an equation (= something) instead of an expression.
- Using the wrong variable or not defining the variable used.

Q7 | Absolute Value Inequalities (SPreCalc8M 1.8)

Points: 2 pts

Part (algebraic solution)

- 1 pt — Correctly rewrites $|A| < c$ as $-c < A < c$ (or $|A| > c$ as $A < -c$ OR $A > c$ for the $>$ case).
- Solves the resulting compound inequality/inequalities for x with no arithmetic errors.

Part (interval notation / graph)

- 1 pt — States the answer in correct interval notation matching the algebraic solution.
- OR draws correct graph: open circles at endpoints (for strict inequalities), shaded region in between (for $<$) or outside (for $>$).

Common Errors (deduct points):

- Using $\leq / \geq$ symbols when the original inequality is strict ($< / >$).
- Forgetting to flip the inequality sign when dividing by a negative number.
- Writing union instead of intersection for a $<$ inequality (or vice versa for $>$).

Q8 | Distance & Midpoint (SPreCalc8M 1.9)

Points: 3 pts

Part (a – Plot)

- 1 pt — Both points plotted correctly on the coordinate grid.

Part (b – Distance)

- 1 pt — Applies $d = \sqrt{[(x_1 - x_2)^2 + (y_1 - y_2)^2]}$ with the correct coordinates.
- Arithmetic must be correct; final answer should be simplified (exact or decimal as specified).

Part (c – Midpoint)

- 1 pt — Applies $((x_1 + x_2)/2, (y_1 + y_2)/2)$ with the correct coordinates and simplifies.

Common Errors (deduct points):

- Subtracting coordinates in the wrong order inside the square root (does not affect distance, but watch for sign errors that propagate).
- Forgetting to take the square root for distance.
- Adding instead of averaging for the midpoint.

Q9 | Linear Functions from Graphs (SPreCalc8M 2.5)

Points: 2 pts

■ **CORRECTION FROM ORIGINAL SOLUTION KEY:** The solution key used the x-intercept to back-solve for b. This is mathematically valid but error-prone. The standard approach — and what should be assessed here — is to read the **y-intercept directly from the graph** (where the line crosses the y-axis). Accept either method, but flag any solution that misreads the intercept from the graph.

Part (a) — Slope (Rate of Change): 1 pt

- Select two clearly identifiable lattice points on the line.
- Compute slope = $\Delta y / \Delta x$. Sign must be correct (negative slope for a falling line).
- 1 pt for the correct slope value; 0 for the wrong sign or wrong calculation.

Part (b) — Equation $f(x) = ax + b$: 1 pt

- Use the slope found in (a).
- Determine b by reading the y-intercept directly from the graph (where $x = 0$), OR substitute any known point (x, y) into $y = ax + b$ and solve for b.
- 1 pt only if both slope AND y-intercept are correct. Deduct this point if the slope from (a) was wrong and carried forward (do not double-penalize a correct method).
- Accept equivalent forms (e.g., point-slope) if correctly simplified to slope-intercept.

Common Errors:

- Misreading the y-intercept from the graph (most common — award partial credit for correct slope).
- Using the x-intercept as if it were the y-intercept.
- Arithmetic error when substituting a point to solve for b.

Q10 | Domain & Range of Square Root (SPreCalc8M 2.3)

Points: 2 pts

Part (domain)

- 1 pt — Sets the radicand ≥ 0 and solves the resulting inequality correctly.
- Expresses domain in interval notation $[a, \infty)$. Bracket required at the finite endpoint.

Part (range)

- 1 pt — Recognizes that $\sqrt{}$ always outputs non-negative values; range is $[0, \infty)$.
- If the function includes a vertical shift (e.g., $\sqrt{x-5} + 3$), the range shifts accordingly — adjust expected answer.

Common Errors (deduct points):

- Using a strict inequality (open interval) for the domain endpoint.

– Writing the range as $(-\infty, \infty)$ without considering the square root constraint.

Q11 | Table of Values & Graphing (SPreCalc8M 2.2)

Points: 7 pts

Part (table)

- ~5 pts — 1 pt per correct $(x, f(x))$ pair (adjust to actual number of rows in the randomized version).
- Award partial credit: deduct 1 pt per incorrect calculation.

Part (graph)

- ~2 pts — Shape/type of curve is correct (1 pt) AND passes through the plotted points (1 pt).
- Deduct if graph contradicts the computed table values.

Common Errors (deduct points):

- Sign errors when squaring negative x values (e.g., $(-3)^2$ should be 9, not -9).
- Applying a vertical shift incorrectly (e.g., forgetting the -1 in x^2-1).
- Drawing the wrong type of curve (line instead of parabola, etc.).

Q12 | Net Change & Average Rate of Change (SPreCalc8M 2.4)

Points: 2 pts

Part (a – Net change)

- 1 pt — Computes $h(b) - h(a)$ correctly with correct sign.
- Deduct if subtraction order is reversed without acknowledgment, or if sign of result is wrong.

Part (b – Average rate of change)

- 1 pt — Divides net change by $(b - a)$. Formula: $[h(b) - h(a)] / (b - a)$.
- This is the slope of the secant line. Sign and magnitude must both be correct.

Common Errors (deduct points):

- Dropping a negative sign when subtracting $h(a)$ — most common error.
- Dividing by the wrong quantity (using $a-b$ instead of $b-a$, flipping the sign).
- Computing $f(b) - f(a)$ correctly but then dividing by the wrong interval width.

Q13 | Function Transformations (SPreCalc8M 2.6)

Points: 1 pt

Correct Method:

- Identify the parent function and all transformations applied (horizontal shift, vertical shift, reflection, stretch/compression).
- Apply transformations in the correct order.
- Key rule: replacing x with $(x + h)$ shifts LEFT by h ; $(x - h)$ shifts RIGHT by h .
- 1 pt for a correctly transformed graph with the new vertex/key point in the right location.

Common Errors (deduct points):

- Shifting in the wrong direction (right instead of left for $(x+h)$).
- Confusing horizontal and vertical shifts.
- Correct shift direction but wrong magnitude.

Q14 | Function Composition (SPreCalc8M 2.7)

Points: 2 pts

Part (a – $(f \circ f)(\text{value})$)

- 1 pt — Evaluates the INNER function first, then substitutes that result into the OUTER function.
- Both steps must be shown or clearly implied; correct final value required.

Part (b – $(g \circ g)(\text{value})$)

- 1 pt — Same process: evaluate g at the input first, then evaluate g again at that result.
- Most common mistake: computing $g(\text{input} - 1)$ instead of $g(g(\text{input}))$. Watch for this.

Common Errors (deduct points):

- Multiplying the two functions instead of composing them.
- Applying the outer function to the original input rather than to the inner function's output.
- Arithmetic error in the squaring step (forgetting to square, or squaring incorrectly).

Q15 | Inverse Functions from a Table (SPreCalc8M 2.8)

Points: 1 pt

Correct Method:

- For a one-to-one function: $f^{-1}(f(x)) = x$ for all x in the domain.
- Apply this identity directly: the answer equals the original input value.
- If the student prefers to verify via the table: look up $f(\text{input}) = \text{output}$, then find the x for which $f(x) = \text{output}$. Must arrive at the correct value.
- 1 pt for the correct numerical answer with any valid reasoning.

Common Errors (deduct points):

- Looking up the wrong table entry (reading row instead of column, or vice versa).
- Confusing $f^{-1}(x)$ with $1/f(x)$.

Q16 | Local Extrema of Polynomials (SPreCalc8M 3.2)

Points: 2 pts

Part (local max)

- 1 pt — Correctly identifies the local maximum value(s) and the x -value(s) where they occur, OR correctly states 'none'.

Part (local min)

- 1 pt — Correctly identifies the local minimum value(s) and the x -value(s) where they occur, OR correctly states 'none'.

Common Errors (deduct points):

- Confusing local and absolute (global) extrema.
- Reporting the x -value when the y -value is asked for (or vice versa).
- Missing an extremum when the graph has multiple turning points.

Q17 | Polynomial Long/Synthetic Division (SPreCalc8M 3.3)

Points: 1 pt

Correct Method:

- Write out ALL coefficients in descending order of degree, inserting 0 for any missing-degree terms.
- Perform synthetic division (or polynomial long division) correctly.
- Express result as: Dividend = Divisor $\times$ Quotient + Remainder.
- 1 pt for completely correct quotient polynomial AND remainder; 0 if either is wrong.

Common Errors (deduct points):

- Omitting zero placeholder coefficients for missing degree terms — this is the most common mistake.
- Arithmetic error in one row of the synthetic division table (propagates through all later rows).
- Stating the quotient without the correct remainder.

Grader Notes:

If student uses long division instead of synthetic — full credit for a correct result.

Verify by expanding: Divisor $\times$ Quotient + Remainder should equal the original polynomial.

Q18 | Polynomial Inequalities (SPreCalc8M 3.7)

Points: 1 pt

Correct Method:

- Move all terms to one side to get polynomial expression < 0 (or > 0).
- Factor completely — find all real roots (critical points).
- Perform sign analysis: test a value in each interval between critical points.
- Write the solution as a union of intervals; use open intervals for strict inequalities.
- 1 pt only for a completely correct interval notation answer.

Common Errors (deduct points):

- Forgetting to move all terms to one side first.
- Incorrect factoring (especially when leading coefficient $\neq 1$).
- Sign error in one interval flipping the entire solution.
- Using closed brackets at critical points for a strict inequality.

Q19 | Exponential Transformations (SPreCalc8M 4.1)

Points: 4 pts

Part (graph)

- 1 pt — Correct shape (increasing or decreasing exponential) AND correct direction of vertical shift.
- Deduct if asymptote is drawn at $y = 0$ instead of the shifted position.

Part (domain)

- 1 pt — $(-\infty, \infty)$ for any exponential function (no restriction on input).

Part (range)

- 1 pt — Correct interval reflecting the vertical shift; open bracket at the asymptote value.

Part (asymptote)

- 1 pt — Horizontal asymptote at $y =$ (vertical shift value).
- If the function is $b^x + k$, asymptote is $y = k$.

Common Errors (deduct points):

- Keeping the asymptote at $y = 0$ after a vertical shift.
- Closing the range interval at the asymptote value (should be open parenthesis).
- Reflecting the graph when there is no negative sign (or missing a reflection when there is one).

Q20 | Logarithm $\leftrightarrow$ Exponential Form (SPreCalc8M 4.3)

Points: 2 pts

Part (a)

- 1 pt — Correctly applies: $\ln(A) = B \blacksquare A = e^B$. Both sides of the converted equation must be correct.

Part (b)

- 1 pt — Same conversion rule applied to a different equation. Correct exponential form required.

Common Errors (deduct points):

- Writing $e \times B$ instead of e^B .

- Converting in the wrong direction (exponential $\rightarrow$ log when log $\rightarrow$ exponential is asked).
- Confusing log base 10 with $\ln$ (natural log, base e).

Q21 | Laws of Logarithms (SPreCalc8M 4.4)

Points: 1 pt

Correct Method:

- Apply the appropriate log law (product, quotient, or power rule) to simplify.
- Quotient rule: $\log_b(A) - \log_b(B) = \log_b(A/B)$.
- Product rule: $\log_b(A) + \log_b(B) = \log_b(A \cdot B)$.
- Power rule: $\log_b(A^n) = n \cdot \log_b(A)$.
- Evaluate to a numeric answer if possible (e.g., $\log_b(b^k) = k$).
- 1 pt for correct application and simplification to a final numeric or simplified log value.

Common Errors (deduct points):

- Adding arguments instead of multiplying ($\log A + \log B \neq \log(A+B)$).
- Using the wrong base when evaluating.

Q22 | Inverse Trigonometry (SPreCalc8M 5.5)

Points: 1 pt

Correct Method:

- Step 1: Evaluate the inner inverse trig function — identify the angle θ in its principal range.
- Step 2: Evaluate the outer trig function at that angle θ .
- Remember principal ranges: $\sin^{-1} \in [-\pi/2, \pi/2]$, $\cos^{-1} \in [0, \pi]$, $\tan^{-1} \in (-\pi/2, \pi/2)$.
- 1 pt for the correct numeric answer.

Common Errors (deduct points):

- Evaluating the outer function at the original argument instead of at θ .
- Using degrees instead of radians (or vice versa) inconsistently.
- Incorrect principal-value angle for the inverse function.

Q23 | Sine Function Transformations (SPreCalc8M 5.3)

Points: 3 pts

Part (graph)

- 1 pt — Correct shape reflecting all transformations (amplitude, reflection, phase/vertical shift).

Part (domain & range)

- 1 pt — Domain $(-\infty, \infty)$; Range correctly reflects amplitude and any vertical shift.

Part (amplitude & period)

- 1 pt — Amplitude = $|A|$ (absolute value of the coefficient); Period = $2\pi/|B|$ for $A \cdot \sin(Bx)$.
- A negative coefficient reflects the graph but does NOT change the amplitude.

Common Errors (deduct points):

- Listing a negative amplitude (amplitude is always positive — it's a magnitude).
- Confusing amplitude and period.
- Incorrect period when $B \neq 1$.

Q24 | Period of Cotangent (SPreCalc8M 5.4)

Points: 2 pts

Part (period)

- 1 pt — Uses the CORRECT formula for cotangent/tangent: $\text{Period} = \pi / |B|$.
- NOT $2\pi/|B|$ — that formula is for sine and cosine only.
- The coefficient A (vertical stretch) does NOT affect the period.

Part (graph / asymptotes)

- 1 pt — Correctly identifies the location and spacing of vertical asymptotes, OR graphs one full period correctly.

Common Errors (deduct points):

- Using $2\pi/|B|$ (the sine/cosine formula) instead of $\pi/|B|$ — award 0 for the period part.
- Including the vertical stretch coefficient A in the period calculation.

Grader Notes:

This is one of the most commonly missed questions. Be strict: the wrong formula means 0 for that sub-point.

Q25 | Vector Representation (SPreCalc8M 8.5)

Points: 1 pt

Correct Method:

- Add the vector's components to each given initial point to find the terminal point.
- All arrows must be the same length, same direction (parallel and equal vectors).
- Optional: compute magnitude using $\sqrt{a^2 + b^2}$.
- 1 pt for correctly plotting all required vectors (same direction and magnitude from each starting point).

Common Errors (deduct points):

- Subtracting instead of adding the vector components.
- Drawing arrows of different lengths from each starting point.
- Reversing the direction of the vector (pointing opposite way).

General Grading Guidelines

Partial Credit: Award partial credit within multi-part questions as described above. Do not penalize the same error twice (carry-forward errors).

Equivalent Forms: Accept any mathematically equivalent form of the answer unless the question specifies a particular form (e.g., interval notation, slope-intercept form).

Unsimplified Answers: Deduct credit if the question asks to 'simplify' and the student's answer is not fully simplified (e.g., leaving $\frac{4}{8}$ instead of $\frac{1}{2}$).

Crossed-Out Work: Grade only what is circled or clearly indicated as the final answer. Ignore crossed-out work.

Q9 Correction: The original solution key used an x-intercept back-substitution to find b. The correct and expected approach is to read the y-intercept directly from the graph. Accept both methods; flag graph misreads.

Questions with Graphs: For any question requiring a graph (Q7, Q8, Q9, Q11, Q13, Q19, Q23, Q24), grade the sketch generously on shape but strictly on key features (intercepts, vertices, asymptotes, direction).

Rubric prepared for randomized versions of Precalculus Placement Self-Assessment 1 (SPreCalc8M). Total: 50 points. Passing threshold and grade cutoffs to be set by the administering instructor.